

A320 takeoff calculator and MCDU uplink takeoff data

The flyPad takeoff screen of the A380, with the FBW A320neo performance model, sending its results to the MCDU as uplink takeoff data (FCOM DSC-22_20-50-10-28)

Aircraft	A32NX (the A380 part of the screen is in the PR 11 manual)
Where to find it	flyPad > Performance > Takeoff ; MCDU > PERF TAKE OFF > <TO DATA (6L)
PR branch	pr/12-a320-takeoff-calculator (stacked on PR 11)
Status	The flyPad calculation was used in the sim; the MCDU uplink pages are checked in the harness; the full flow is still to re-test in the sim.
Pictures	flyPad and MCDU in the local browser harness, running the build installed in the sim. The harness has no navigation database and no flight plan: the runway is typed and the uplink is sent to the MCDU by the harness (the same message the flyPad sends).

1. What it is for
2. The flyPad screen (A320)
3. Sending to the MCDU
4. What to expect
5. Limitations and why

1. What it is for

The A320 takeoff speeds, FLEX temperature and THS come from the airline or the crew's own tool. Here the flyPad computes them with the FlyByWire A320neo performance model (built from Airbus data) and sends them to the FMGC as an **uplink takeoff data** message, which the crew inserts from the MCDU PERF TAKE OFF page as on the real aircraft.

2. The flyPad screen (A320)

The screen is the one of the A380 (see the PR 11 manual) with these A320 differences:

- No noise procedure fields; T.O CG is optional: below 27 % MAC the forward CG performance applies (A320 FCOM PER-LOD) and the THS is shown in RESULTS.
- All values come from the FBW A320 model: they are shown in green (no estimates).
- Wet and contaminated runways are computed (the A380 page accepts dry only).

Mon Sep 28 1024Z 100%

Performance

Takeoff Top of Descent Landing Temperature Correction

Airport ICAO Fill data from FMS

RUNWAY

Runway Manual

Heading °

TORA m m

Elevation ft

Slope %

Entry angle 90°

CONDITIONS

Condition Dry

Wind °/kt

OAT °C °C

QNH hPa hPa

Anti-Ice Off

Packs On

AIRCRAFT

TOW t t

T.O CG % MAC

Flaps CONF 1+F

Thrust FLEX

DEPARTURE

THR RED ft

ACCEL ft

EO ACCEL ft

RESULTS

TOW --- t MTOW(perf) --- t

V1 --- kt MARGIN --- m

VR --- kt

V2 --- kt FLAPS -

T.O CG --- % T.O SHIFT --- m

THR RED --- ft ACCEL --- ft

EO ACCEL --- ft

Weights, speeds, FLEX and THS: FlyByWire A320neo performance model (from Airbus data).

Calculate Clear Send to FMS

TAKEOFF RUN

Thrust TOGA FLEX - --- +

Calculate to see the takeoff run

Required: FlyByWire A320neo performance model (all limitations). V1, VR, 35 ft: estimates, all engines.

Figure 1. The A320 takeoff screen before any input.

Mon Sep 28 1024Z 100%

Performance

Takeoff Top of Descent Landing Temperature Correction

Airport ICAO Fill data from FMS

RUNWAY

Runway Manual

Heading 143

TORA 3500 m m

Elevation 499

Slope 0

Entry angle 90°

CONDITIONS

Condition Dry

Wind 140/8

OAT 22 °C °C

QNH 1015 hPa hPa

Anti-Ice Off

Packs On

AIRCRAFT

TOW 68 t t

T.O CG % MAC

Flaps CONF 1+F

Thrust FLEX

DEPARTURE

THR RED 2000

ACCEL 2000

EO ACCEL 2000

RESULTS

TOW 68.0 t MTOW(perf) 86.5 t

V1 148 kt MARGIN 519 m

VR 148 kt FLEX 64°C

V2 148 kt FLAPS 1+F

T.O CG --- % T.O SHIFT 0 m

THR RED 2000 ft ACCEL 2000 ft

EO ACCEL 2000 ft

Weights, speeds, FLEX and THS: FlyByWire A320neo performance model (from Airbus data).

Calculate Clear Send to FMS

TAKEOFF RUN

Required 2,961 m Available 3,480 m Margin 519 m

Thrust TOGA FLEX - 64°C + 44° 64° max

35 ft 2,575 m

VR 2,243 m

V1 2,243 m

Required 2,961 m

Required: FlyByWire A320neo performance model (all limitations). V1, VR, 35 ft: estimates, all engines.

Figure 2. LFBO 14R-like runway (3500 m, 499 ft), 68 t, 22 °C: FLEX 64 °C (maximum), required 2961 m of 3480 m, margin 519 m.

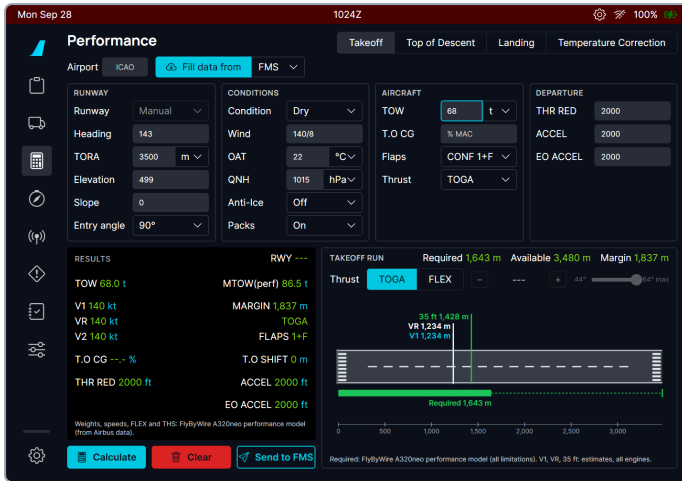


Figure 3. The TOGA run at the same weight: 1643 m.

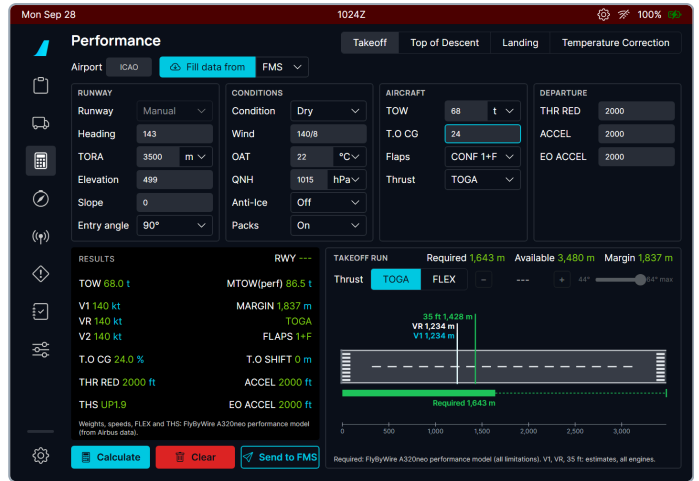


Figure 4. T.O CG 24 % (forward CG) at TOGA: the THS (UP1.9) appears in RESULTS.

NOTE

At light weights the FBW A320 model gives **V1 = VR = V2** (the pictures show 148/148/148 at FLEX and 140/140/140 at TOGA for 68 t). This comes from the FlyByWire model itself (the same with its version before this PR), not from this feature.

3. Sending to the MCDU

1. On the flyPad, choose the **runway from the list** (the uplink carries the runway ident), calculate, choose TOGA or the FLEX temperature for the takeoff run, and press **Send to FMS**.
2. The flyPad says "Takeoff data sent to the FMS: MCDU PERF TAKE OFF, UPLINK TO DATA, RECEIVED TO DATA", and warns right away if the FMGC departure runway or TOW will prevent the insertion.
3. On the MCDU the scratchpad shows **TAKEOFF DATA UPLINK**.
4. PERF TAKE OFF > <**TO DATA** (6L) > **RECEIVED** <**TO DATA** (6L): the UPLINK MAX TO DATA page (or FLX TO DATA with 4R). Check it, then **INSERT UPLINK**.

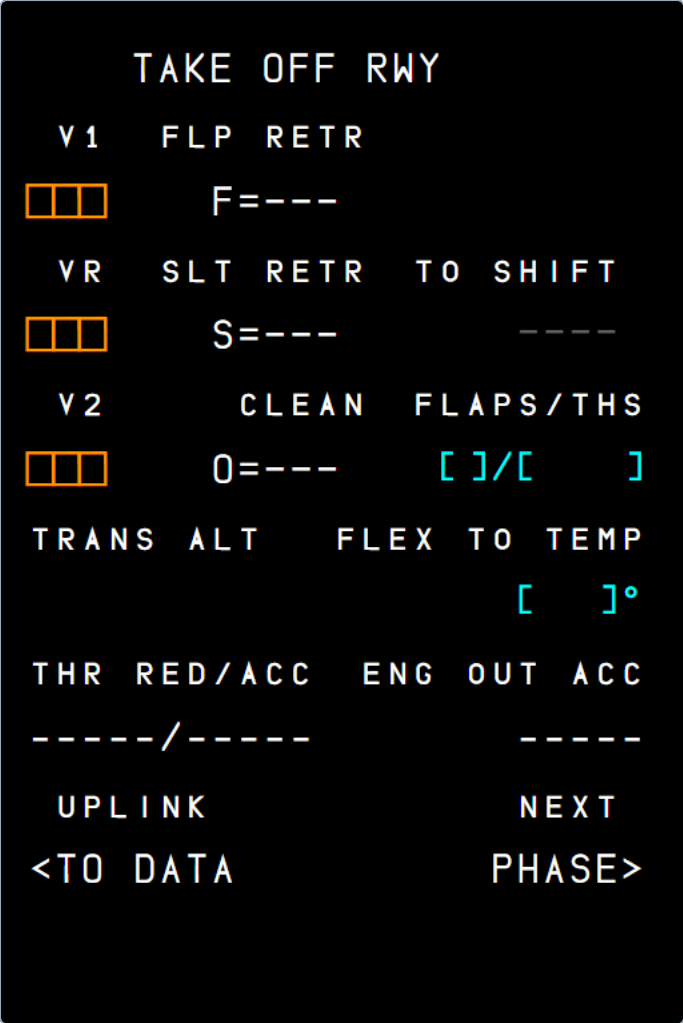


Figure 5. PERF TAKE OFF: the new UPLINK <TO DATA prompt (6L), in PREFLIGHT and DONE.

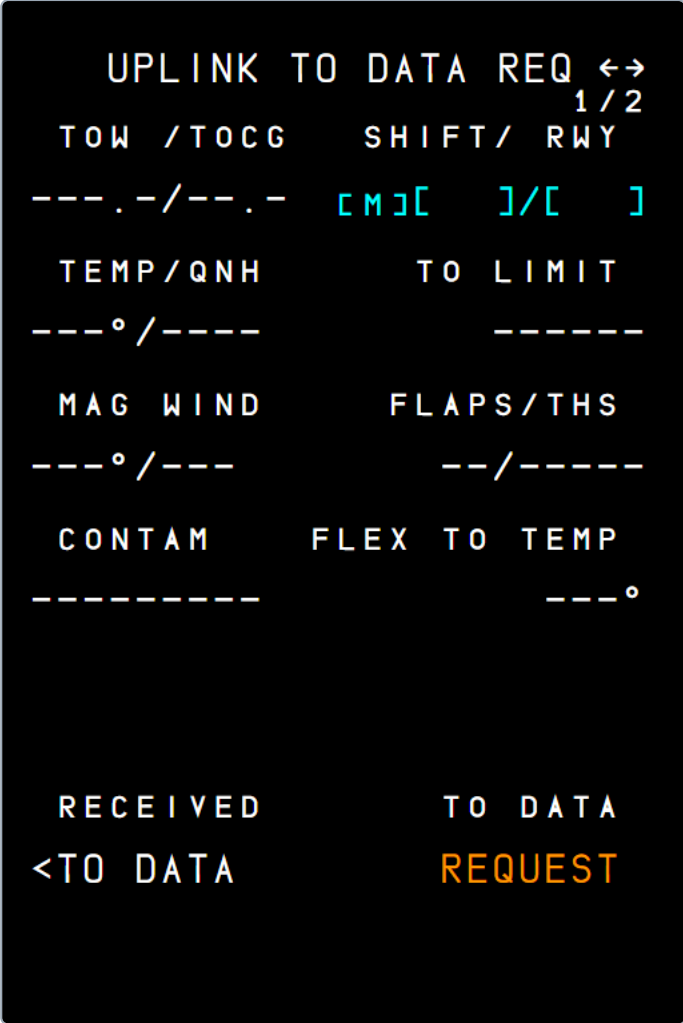


Figure 6. UPLINK TO DATA REQ: the request page (two runways, 1/2). RECEIVED <TO DATA (6L), TO DATA REQUEST (6R).

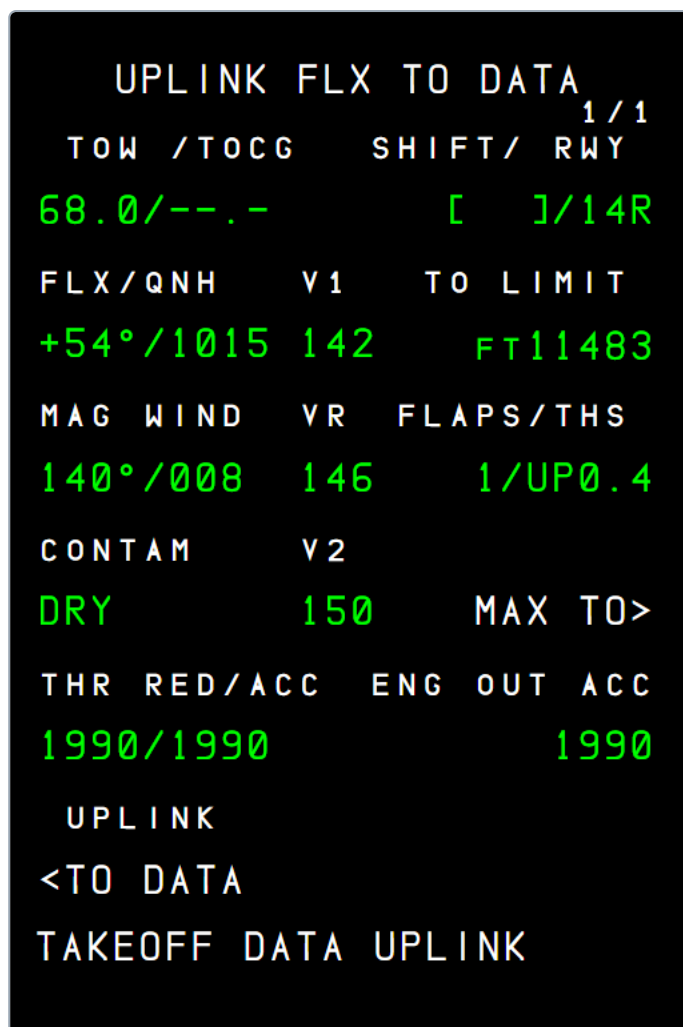
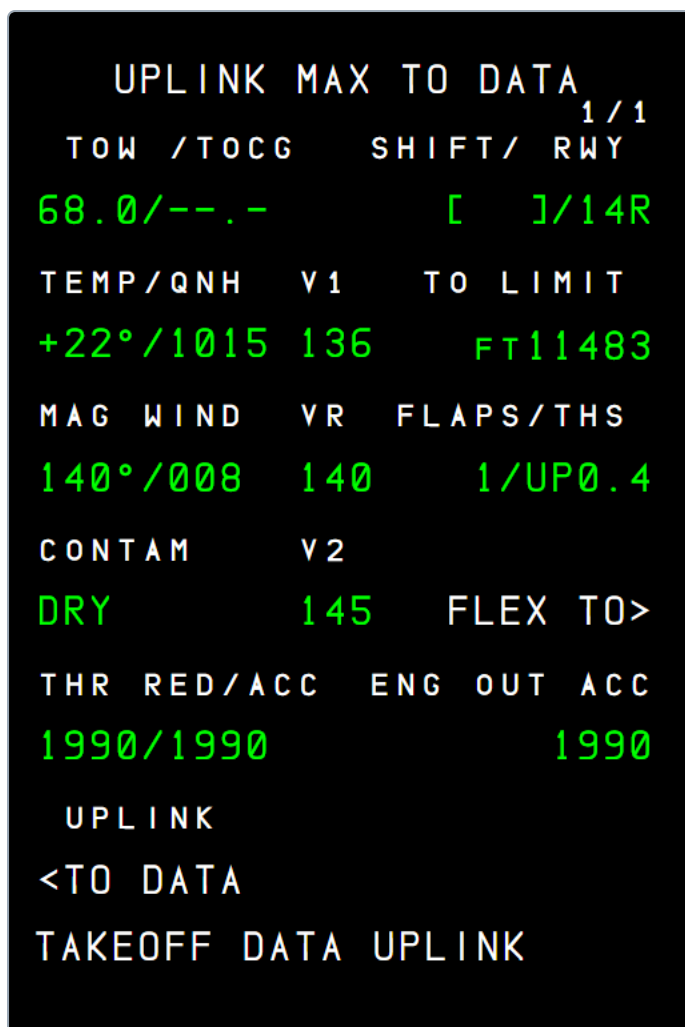


Figure 7. UPLINK MAX TO DATA after TAKEOFF DATA UPLINK: TOW, runway 14R, temperature and QNH, V1 / VR / V2, TO LIMIT, flaps / THS, THR RED / ACC, ENG OUT ACC. FLEX TO> (4R) shows the FLEX data.

Figure 8. UPLINK FLX TO DATA: FLX +54 °C with its own speeds.

There is **no INSERT UPLINK prompt** in these pictures because the harness FMGC has no flight plan. The prompt appears only when the uplink is complete (V1 and VR present) and the checks below pass.

The insertion rules (A320 FCOM)

- The uplink runway must be the departure runway of the flight plan.
- The uplink TOW must be **not more than 1 t below or 3 t above** the FMGC TOW; otherwise there is no insertion.
- INSERT UPLINK fills V1, VR, V2, FLAPS/THS, FLEX TO TEMP (cleared for a MAX takeoff), THR RED/ACC, ENG OUT ACC and the TO SHIFT.
- An incomplete uplink (for example without V1 or VR) gives no INSERT UPLINK prompt (DSC-22_20-70 P 6).



Figure 9. An uplink the FMGC cannot read: INVALID TAKEOFF UPLINK in the scratchpad.

MCDU message	Meaning
TAKEOFF DATA UPLINK	Uplink takeoff data received.
INVALID TAKEOFF UPLINK	The uplink is not valid (missing data).
REQUEST IS PENDING	A request was sent and has no answer yet.
NO ANSWER TO REQUEST	No answer 4 minutes after the request.

PERF TAKE OFF also gets the **TO SHIFT** field (1 m up to the runway length, CLR to delete): the distance lost when taking off from an intersection.

4. What to expect

Situation	What you should see
Send to FMS with a runway from the list	Success message; TAKEOFF DATA UPLINK on the MCDU.
Send to FMS with a manual runway	Nothing is sent: choose the runway in the list.
FMGC departure runway differs	flyPad warning; no INSERT UPLINK on the MCDU.
FMGC TOW differs by more than -1 t / +3 t	flyPad warning "INSERT UPLINK needs a TOW within -1 t / +3 t"; no INSERT UPLINK.
No TOW in the FMGC	flyPad warning "No TOW in the FMS (INIT B: ZFW and BLOCK)": fill INIT B first.
INSERT UPLINK	PERF TAKE OFF filled with the uplink.

5. Limitations and why

Limitation	Why	What to do
Only the thrust selected for the takeoff run is sent.	The airline sends what the crew asked for; the MCDU holds MAX and FLEX pages, the flyPad fills the one you use.	Select TOGA or FLEX before sending.
$V1 = VR = V2$ at light weights.	The FlyByWire A320 model (upstream).	Check against your own tools.
No real airline datalink.	The flyPad plays the airline.	-